INTERIM FEDERAL SPECIFICATION

DETERGENT-SANITIZER-GENERAL PURPOSE, SUPER CONCENTRATE

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION.

1.1 Scope. This specification covers a plastic pouch-pac containing 12 ounces (by weight) of a liquid super concentrate detergent-sanitizer, which when diluted to a U. S. gallon with water, will provide a cleaner-sanitizer, concentrate intended for general purpose building maintenance.

1.2 Classification.

1.2.1 Type and grade. This specification covers one type and grade of detergent-sanitizer.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS.

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bid, form a part of this specification:

Federal Specifications:

PPP-B-636 - Box, Fiberboard.

Federal Standards:

Fed. Std. No. 102 - Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specifications:

MIL-B-121B - Barrier Material, Greaseproofed, Waterproofed, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

Governmental:

Federal Fungicide, Insecticide, and Rodenticide Act.

(Copies of the above Act may be obtained from the Superintendent of Documents, Government Printing Office, Washington, D. C.)

Official Methods of Analysis of the Association of Official Agricultural Chemists.

(Copies of the above document may be obtained from the Association of Official Agricultural Chemists, P. O. Box 540, Benjamin Franklin Station,

American Society for Testing and Materials (ASTM) Standards:

D 882- Methods of Test for Evaluating Treated Textiles for Permanence of Resistance to Microorganisms.

D 1505 - Methods of Test for Density of Plastics by the Density-Gradient Technique.

D 1709 - Method of Test for Impact Resistance of Polyethylene Film by the Free Falling Dart Method

E 96 - Method of Test for Water Vapor Transmission of Materials in Sheet Form.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Official and Tentative Methods of the American Oil Chemists Society

Da2b-42 - Soap and Soap Products - Moisture Distillation Method
Db2-48 - Soap Containing Synthetic Detergents, Alcohol soluble and Insoluble Matter.
Da23-56 - Soap and Soap Products, Free Glycerol

(Copies of the above methods may be obtained from the American Oil Chemists Society, 35 E. Wacker Drive, Chicago 1, Illinois.)

3. REQUIREMENTS.

3.1 Material. The product shall be a homogeneous liquid consisting of anionic and non ionic detergents combined with suitable chelating, and sequestering agents and sanitizer to provide an effective cleaner-sanitizer when diluted to working solution strength. It shall be effective as a cleaner-sanitizer used with all types of water hardness.

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Table I. Composition

	Percent		Test method
	Minimum	Maximum	
Active anionic detergent*	30		4.4.2
Active nonionic detergent	40		4.4.3
Total volatile		13.5	4.4.4
Total ethers, glycols, polyglycols		6	4.4.9
Total inorganic salts		2	4.4.5
Active tetra sodium ethylenediamine tetra acetate	0.6	-	4.4.6
Aryl-, Chloro aryl and Chloro-benzyl phenol	5.0		

*The manufacturer must furnish the chemical structure and the molecular weight. (If average so specify.)

3.2 Registration. The product shall be registered under the provisions of the Federal Insecticide, Fungicide, and Rodenticide Act. (The product shall kill staphlococcus aureus and salmonella typhosa at a dilution of 1 ounce of super concentrate in one gallon of water when tested in accordance with A.O.A.C. use dilution method (p-65-66 5) - Disinfectants - Official Methods of Analysis A.O.A.C., 9th Edition, 1960).)

3.3 Excluded ingredients. The product shall contain no gum or other hydrocoloid stabilizers, ethoxylated or propoxylated phenols.

3.4 The superconcentrate shall readily disperse in hot tap water to a homogeneous solution.

3.5 Stability. The plastic pouch-pac and superconcentrate must maintain its chemical stability for a period of one year after receipt and show no tendency toward precipitation or sedimentation or leakage while in storage or transit under extreme ranges of temperature (-25°F to 115°F).

3.6 Odor. At use dilution concentrations the product shall be mildly perfumed.

3.7 pH value. The pH of a 1 percent solution of the compound in distilled water shall not be less than 7 or more than 9.5.

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3.8 The film material used to package the superconcentrate cleaner shall meet the following requirements specified in table II.

Table II. Package film material requirements

Property	Requirement	Test method
Film density	0.936 to 0.940 g/cc	Table III
Film ultimate tensile strength	2500 psi (min)	Table III
Film elongation	300 percent (min)	Table III
Water vapor transmission	1.0 grams/100 sq.in./ 24 hrs. (max)	Table III
Greaseproofness	5-1/2 hours minimum (hours to fail)	Table III
Burst strength	2 inch diam. dart, 60 inch fall, 220 grams (50% failure weight)	Table III

3.9 Accelerated storage test. The finished product (plastic pac with superconcentrate) shall show no leakage, breakdown or other deterioration when stored in a forced air oven at 75° C for seven days.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein. For purposes of sampling, an inspection lot for acceptance inspection and tests shall consist of all material of the same type and grade submitted for inspection and delivery at one time.

4.3 Inspection level for acceptance. The inspection level and the acceptable AQLs for the packaging and marking requirements of this specification shall be in accordance with the provisions set forth in MIL-STD-105, level S-2, AQL 4.0 percent defective.

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4.4 Test procedures.

4.4.1 Film material. The film material used to package the product shall conform to the requirements specified in table II, when tested in accordance with the test methods as specified in table III.

Table III

Property	Test method
Film density	ASTM D 1505
Film ultimate tensile strength	ASTM D 882
Film elongation	ASTM D 882
Water vapor transmission	ASTM E 96 (Procedure E)
Greaseproofness	MIL-B-121b, Barrier Material, Greaseproofed, Waterproofed, Flexible
Burst strength	ASTM D 1709 (Method B)

4.4.2 Anionic activity.

4.4.2.1 Reagents.

1. Carbon Tetrachloride C. P.
2. SDA-3A Alcohol (neutralized using O-cresol red as indicator)
3. 13.6% p-toluidine hydrochloride (Eastman reagent grade)
4. Distilled H_2O
5. O-Cresol red indicator
6. 0.1 N NaOH

4.4.2.1.1 Procedure.

1. Into a 250 ml separatory funnel, put 25 ml ethyl alcohol and 50 ml H_2O.
2. Add approximately 3.0 - 5.0 grams sample to separatory funnel from weighing bottle (determine weight by difference), dissolve sample.
3. Add to separatory funnel, 50 ml of carbon tetrachloride and 25 ml of a 13.6% p-toluidine hydrochloride solution.
4. Shake funnel approximately 10 minutes. (Caution: from time to time, release pressure.) Then allow layers to separate.
5. Draw off bottom layer (carbon tet layer) into 500 ml flask, containing 125 ml of previously neutralized ethyl alcohol using O-cresol red as indicator. Be absolutely sure not to draw off any of the top layer.
6. Add 25 ml of carbon tetrachloride to separatory funnel, shake for 2 minutes, allow to separate, and again draw off bottom layer.
7. Repeat step 6 for the third washing.
8. Titrate together all 3 extractions with 0.1 N NaOH to the O-cresol red end point (purple).

Calculation: % Active = $\frac{VXNXE_q.W}{SX10}$

Where: V = Volume (in CC) of NaOH used
N = Normality of NaOH solution
S = Weight of sample

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4.4.3 Non Ionic activity. Take 10 gram sample (accurately weighed), dry overnight at 50°C and extract with at least three 50 ml portions of acetone (extraction should be made at approximately 50°C with vigorous stirring. Evaporate the extra to constant weight at 50°C in a tared beaker. Calculate the percent non ionics as follows:

$$\text{Activity, percent} = \left(\frac{A}{B} \times 100\right) - (C \neq D)$$

Where: A = residue weight
B = sample weight
C = percent anionics (see 4.4.2).
D = percent glycols (see 4.4.9).

4.4.4 Total volatile. Total volatile shall be determined in accordance with A.O.C.S. Method Da2b-42.

4.4.5 Inorganic salts. Total inorganic salts shall be determined in accordance with A.O.C.S. Method Db 2-48.

4.4.6 Tetra sodium ethylenediamine tetra acetate (EDTA) determination.

4.4.6.1 Reagents.

1. Potassium chromate solution.

   Weigh out 2.5 grams of potassium chromate, dissolve in distilled water, and dilute to 500 ml in a volumetric flask.

2. EDTA standard solution.

   Weigh out 50, 100, 200, and 300 mg of dried A. R. EDTA into each of four 100 ml volumetric flasks. Add approximately 75 ml of distilled water to each flask and dissolve. Add one NaOH pellet for each 100 mg EDTA and dissolve. Dilute flasks to 100 ml with distilled water.

3. Arsenious acid, 0.2 N.

   Weigh out 4.93 gm of arsenious acid. Add 3.5 gm of NaOH, and dissolve in about 150 ml of distilled water (warm). Neutralize with glacial acetic acid using phenolphthalone indicator, dilute to about 270 ml and slowly add 200 ml of glacial acetic acid with thorough mixing. Dilute to 500 ml in volumetric flask.

4. Methanol - distilled water solvent (1:1).

4.4.6.1.1 Procedure.

Preparation of standard curve:

1. Pipette 2.0 ml of each of the standard solutions into calibrated spectrophotometer tubes.
2. Add 2.0 ml of the potassium chromate solution and mix.
3. Add 2.0 ml of the arsenious acid reducing agent.
4. Place the tubes in a boiling water bath for 5 minutes, then remove and cool.
5. Add 10.0 ml of distilled H_2O methanol mixture (1:1) and shake.
6. Read the absorbance of the colored solution on the spectrophotometer at 560 m/μ
7. Prepare a blank for reference by using 2.0 ml distilled water in place of an EDTA standard and proceed as above.

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Sample analysis:

8. Weigh a 10.00 gm sample into a 100 ml volumetric flask and dissolve with about 60 ml of the distilled water-methanol mixture, dilute to 100 ml with same solvent.

9. Transfer a 10.0 ml aliquot of the sample solution to a small beaker and acidify it with 1.0 ml of glacial acetic acid.

10. Filter the acid solution through No. 42 filter paper to remove any precipitate and colloid and collect the filtrate in a 25 ml volumetric flask.

11. Wash the filter paper well with the methanol distilled water mixture, collecting the washings in the 25 ml flask.

12. Dilute the solution to the mark using methanol-water. Take a portion (15 ml) and add 15 ml of chloroform. Shake well and let stand 1 hour.

13. Pipette a 5.0 ml aliquot of the supernatant liquid into a spectrophotometer tube.

14. Follow steps 2, 3 and 4 above.

15. Add 7.0 ml methanol-distilledwater.

16. Read the absorbance of the colored solution on the spectrophotometer at 560 m_{μ} employing the reference blank used for the standards.

Calculations:

$$\frac{R}{W} \times 100 = \% \text{ EDTA}$$

R = mg. from standard curve obtained from sample absorbance.

W = weight of sample in 5 ml aliquot of step 13. (200 mg.)

4.4.7 Solubility. Add one part of sample to nine parts of hot tap water (115°F) and stir vigorously for one minute. The product should be homogeneous.

4.4.8 pH value. Make a 1.0 percent solution of the compound in freshly boiled and cool (room temperature) distilled water. Determine the pH value of this solution to the nearest 0.1 unit at room temperature (25° C ± 2°C) by means of a suitable pH meter which employs the glass electrode.

4.4.9 Glycerol, Polyglycerol determination. Determine total glycerol and polyglycerols in accordance with A.O.C.S. Method Da23-56 except that the size of the sample in Part D, Item I shall be approximately one gram.

5. PREPARATION FOR DELIVERY

For civil agency procurement the definitions and application of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 Packaging. Twelve ounces (see 1.1) of liquid concentrate shall be packaged in a flexible plastic pouch. The plastic used in fabricating the pouch shall meet the physical requirements of table II. The pouch shall be pillow type with top and bottom seams. Closure of the filled pouch shall be as recommended by the supplier of the material used in fabricating the pouch, except that pressure-sensitive adhesive tapes shall not be used. The plastic bag shall be not less than 3 mil thickness and so constructed and sealed as to insure no leakage during transit or storage. The product shall not affect the package nor the package affect the product.

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5.2 The detergent in the plastic bag shall be unit packaged in a container conforming to PPP-B-636, style OPF except, that the inner flaps shall meet covering the entire length of the container, with the outer flaps meeting at the centers and covering the width of the container. To facilitate closure the outer flaps shall have side tabs. The inner flaps shall have a die-cut opening of sufficient width and length to receive the tabs. The intermediate package shall be suitable for mailing.

5.3 Ten unit packages shall be intermediately packaged in a suitable shipping container in accordance with good commercial practice. The intermediate package of ten shall be overpacked eight cartons to the master carton (master carton will contain 80 units). The outer carton shall be a suitable shipping container in accordance with good commercial practice and shall be suitable for multiple shipping and storage.

5.4 Marking.

5.4.1 Each unit package shall be marked with instructions for dilution and manufacturer's recommended cleaning dilution. (Alternatively these instructions may be printed on an insert label.)

5.4.2 In addition to the labeling specified in 5.4.1 and any special marking required by the contract or order, each unit, intermediate and shipping container shall be marked in accordance with Fed. Std. No. 123.

6. NOTES.

6.1 Intended use. The detergent-sanitizer covered by this specification is intended as a general purpose cleaner-sanitizer for building maintenance.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity of detergent-sanitizer required.

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